A3-PS015-TC001

Explainable Home and Last-Mile Connectivity Diagnosis

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Problem Description

Speed tests hide what matters

Peak download under unloaded conditions ignores packet loss, round-trip latency and loaded jitter, which decide call, game and video quality.

Access media behave differently

Fiber, cellular, fixed wireless and satellite have different latency and loss baselines, so one threshold does not fit all.

Users cannot localise faults

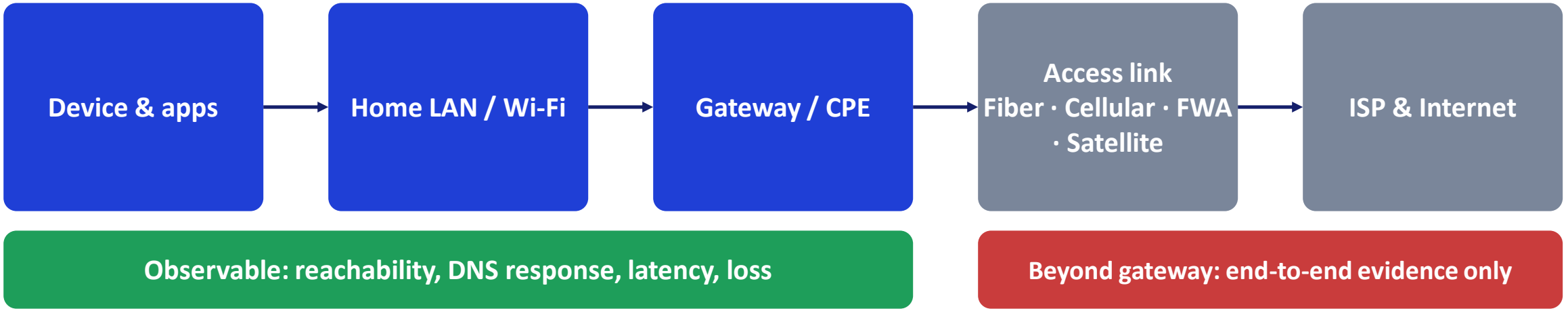
Is it the Wi-Fi, the gateway, DNS or the ISP? Without evidence, people reboot, guess or blame the wrong party.

Tools over-claim

Black-box tools declare cable damage or ISP faults without proof. Trust needs transparent rules and honest uncertainty.

Goal: a guided desktop agent that diagnoses 4 controlled faults with visible evidence and says “Inconclusive” when it cannot isolate the cause

Background: Where Can a Fault Hide?



Speed is not quality

Real-time apps depend on loss, RTT and jitter, not peak bandwidth [1][4][6].

Load reveals problems

Bufferbloat only appears when the link is busy, so we measure latency idle and loaded [6].

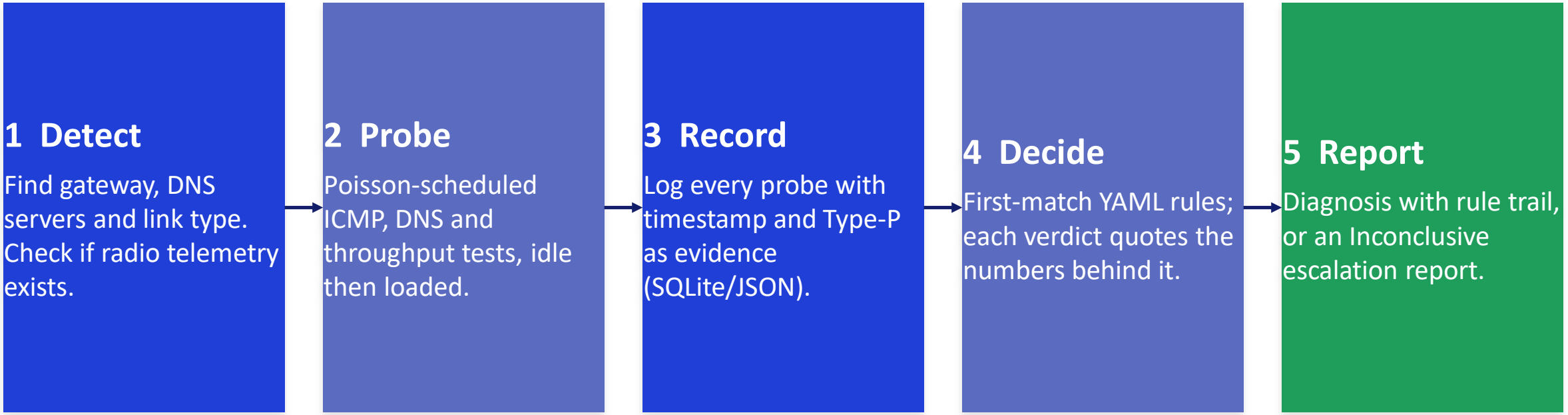
Honest scope

No cable-damage verdicts, no ISP-internal root cause, no Wi-Fi blame without radio telemetry.

Literature & RFC Review

Source	Key idea	How we use it
[1] RFC 2330	IPPM framework: Type-P, singleton, sample, Poisson sampling	Probe design and scheduling
[2][3] RFC 2680 / 2681	One-way loss and round-trip delay metrics	Loss % and RTT definitions
[4] RFC 3393	IP packet delay variation (jitter)	Jitter statistic, idle vs loaded
[5] RFC 7594 (LMAP)	Large-scale broadband measurement framework	Volunteer-endpoint scaling path
[6] Gettys & Nichols	Bufferbloat: queues inflate latency under load	Congestion rule (loaded RTT)
[7] NetEm	Linux network emulation: delay and loss	Controlled fault injection

Proposed Solution: Probe, Decide, Explain



Guardrails built into the rule engine

- No definitive damaged-cable detection
- No ISP-internal root-cause claims; escalate with evidence instead
- No Wi-Fi interference verdict without RSSI / noise / channel telemetry

Explainable Decision-Rule Engine

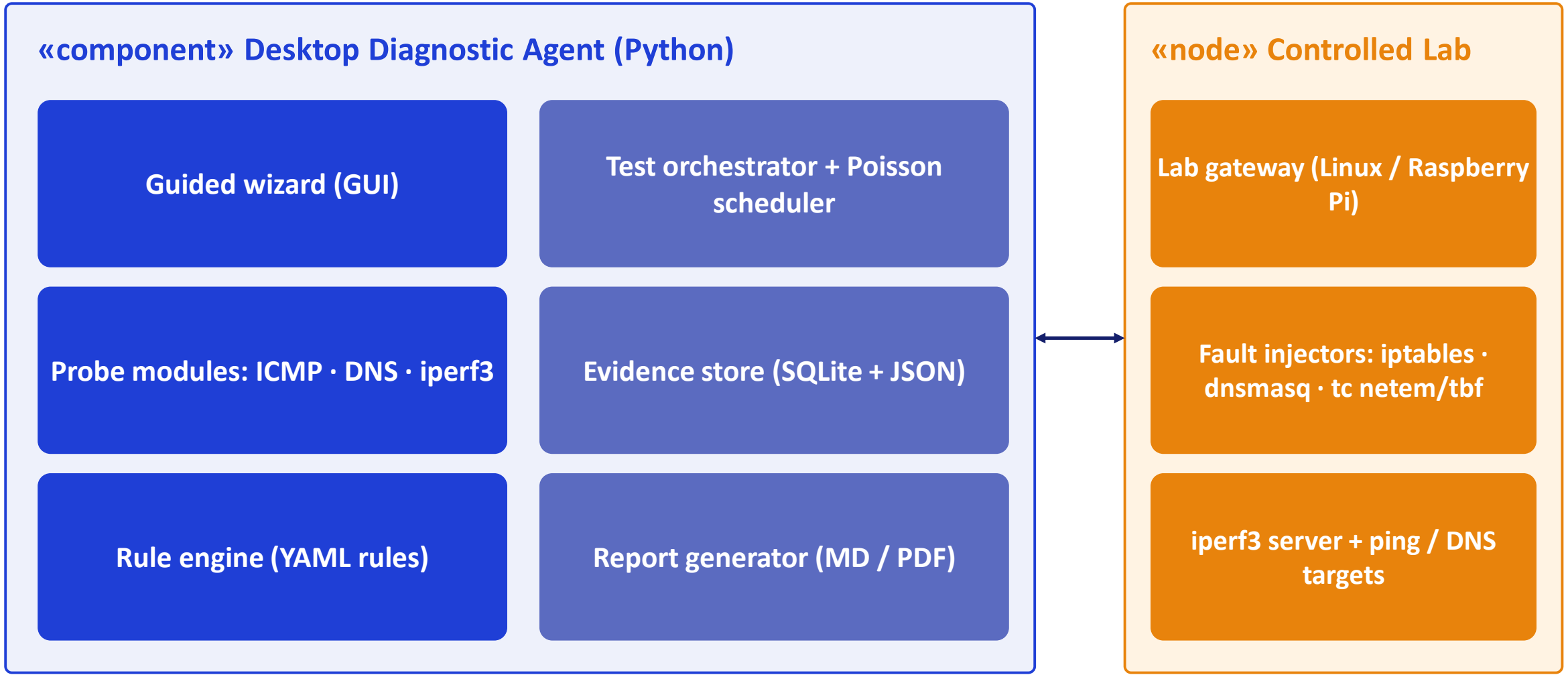
Rule	Evidence required	Verdict
R1 Gateway	0 of N replies from gateway (ICMP and ARP)	Gateway unreachable
R2 DNS	Gateway and IP-literal targets reachable; resolver queries fail	DNS failure
R3 Loss	Idle loss ≥ 3% with normal RTT; located at gateway or beyond	Packet loss
R4 Congestion	Idle RTT normal; loaded RTT ≥ +100 ms or ≥ 3× idle, loss rises	Congestion
R5 Fallback	Conflicting tests, ICMP filtered, or Wi-Fi without radio data	Inconclusive + escalation

Thresholds are starting values, tuned during lab runs. Rules run in order R1 to R5; the first match wins.

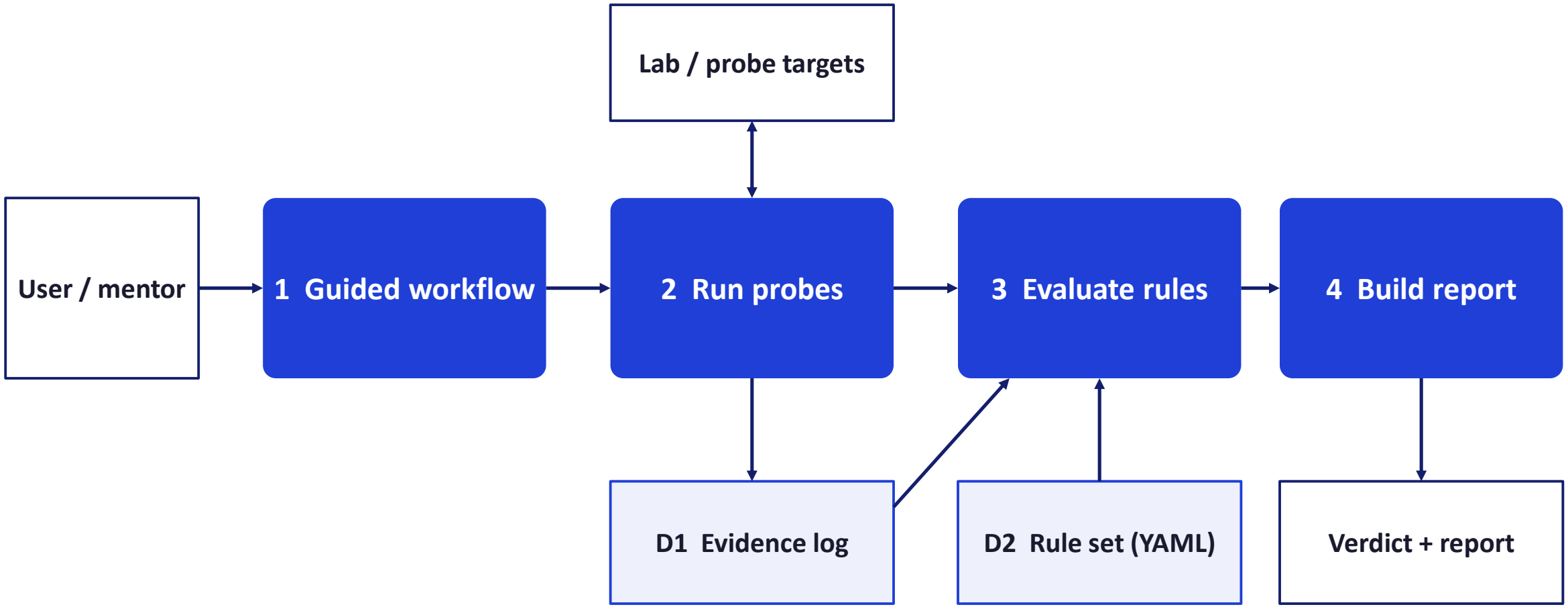
Optimization Proposed: Before vs After

Aspect	Before: speed test	After: proposed agent
Metrics	Peak download only	Latency, loss, jitter, DNS delay, throughput
Load behaviour	Unloaded snapshot	Idle vs loaded latency comparison
Output	One bandwidth number	Verdict with rule trail and evidence
Unknown causes	Guess or silence	Inconclusive + escalation report
Probe timing	Fixed intervals	Poisson sampling per RFC 2330

Solution Architecture & Design



Data Flow Diagram (DFD)



Rectangles: external entities and outputs • Rounded: processes • Shaded boxes: data stores

Evaluation: Four Faults plus Inconclusive Cases

Scenario	Lab injection	Expected verdict	Pass criterion
Gateway unreachable	Drop all traffic to lab gateway (iptables / link down)	Gateway unreachable	≥ 9/10 runs
DNS failure	Stop dnsmasq or block port 53	DNS failure	≥ 9/10 runs
Induced packet loss	tc netem loss 5–10%	Packet loss	≥ 9/10 runs
Network congestion	tc tbf rate cap + iperf3 saturation	Congestion	≥ 9/10 runs
Inconclusive cases	ICMP blocked, mixed faults, Wi-Fi without telemetry	Inconclusive	No over-claim

Reported as: confusion matrix, false-attribution count and a per-case Inconclusive analysis. Targets, not yet measured results.

Deployment Timeline

1 — 2 — 3 — 4

Phase 1: Lab and probes

Build lab
ICMP, DNS, iperf3 probes
Poisson scheduler

Phase 2: Rules and faults

YAML rule engine
4 fault-injection scripts
Guided wizard

Phase 3: Evaluate

Repeated runs per fault
Inconclusive analysis
Tune thresholds

Phase 4: Report and demo

Escalation template
Final report
Live demo + pseudocode

Method: private GitHub repo shared with mentors, small commits, a working increment demoed each phase.

Bibliography

[1] V. Paxson et al., “Framework for IP Performance Metrics,” RFC 2330, IETF, 1998.

[2] G. Almes et al., “A One-way Packet Loss Metric for IPPM,” RFC 2680, 1999.

[3] G. Almes et al., “A Round-trip Delay Metric for IPPM,” RFC 2681, 1999.

[4] C. Demichelis, P. Chimento, “IP Packet Delay Variation Metric for IPPM,” RFC 3393, 2002.

[5] P. Eardley et al., “A Framework for Large-Scale Measurement of Broadband Performance (LMAP),” RFC 7594, 2015.

[6] J. Gettys, K. Nichols, “Bufferbloat: Dark Buffers in the Internet,” Commun. ACM 55(1), 2012.

[7] S. Hemminger, “Network Emulation with NetEm,” linux.conf.au, 2005.

Thank You for Your Attention!

Questions and live demo